ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MATHEMATICAL LITERACY

PAPER 1

MARKS: 100 **TIME: 2 hours**

This question paper consists of 4 pages.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Show ALL calculations clearly. Round off ALL final answers to TWO decimal places, unless stated otherwise.
3. You may use an approved calculator. Indicate units of measurement where applicable.
4. Maps and diagrams are NOT necessarily drawn to scale, unless stated otherwise.

QUESTION 1: FINANCE

Thandeka earns a salary of R6 800 per month. TABLE 1 shows her monthly expenses.

Expense	Amount	Expense	Amount
Rent	R2 500	Savings	R500
Transport	R850	Electricity	R450
Groceries	R1 400	Clothing account	R380
Cellphone	R299		

1.1 Calculate Thandeka's total monthly expenses. (2)

1.2 Calculate how much money she has left over each month. (2)

1.3 Calculate the percentage of her salary that she spends on rent. (3)

1.4 Her rent will increase by 8%. Calculate her new rent. (3)

1.5 Will she still have money left over after the rent increase? Show your calculations. (2)

1.6 Her electricity costs R2,85 per kWh. Calculate the cost of using 158 kWh. (2)

1.7 A school bag costs R345, including 15% VAT. Calculate the price of the bag WITHOUT VAT, and the amount of VAT paid. (3)

1.8 Thandeka needs a new phone. She has two options:
Option A: a contract of R149 per month for 24 months (phone included).
Option B: buy the phone for R1 999 cash and pay R80 per month for prepaid airtime and data.

1.8.1 Calculate the total cost of Option A over 24 months. (2)

1.8.2 Calculate the total cost of Option B over 24 months. (3)

1.8.3 Which option is cheaper, and by how much? (2)

1.9 Give ONE reason why it is important for Thandeka to keep saving every month. (2)

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QUESTION 2: MEASUREMENT

2.1 Convert:

2.1.1 2,5 km to metres (1)

2.1.2 750 mℓ to litres (1)

2.1.3 3 hours 45 minutes to minutes (1)

2.1.4 1 450 g to kilograms (1)

2.2 A rectangular classroom floor is 8 m long and 6,5 m wide.

2.2.1 Calculate the area of the floor. Use: Area = length × width (2)

2.2.2 The floor will be covered with square tiles of 0,5 m × 0,5 m. Calculate how many tiles are needed. (3)

2.2.3 Tiles are sold in boxes of 10 at R189 per box. Calculate the number of boxes needed and the total cost. (4)

2.2.4 Calculate the perimeter of the floor. Use: Perimeter = 2 × (length + width) (2)

2.2.5 Skirting board costs R35 per metre. Calculate the cost of skirting for the whole perimeter. (2)

2.3 A cylindrical water tank has a radius of 0,7 m and a height of 1,5 m. Use: Volume = π × radius² × height, with π = 3,142

2.3.1 Calculate the volume of the tank in m³. (3)

2.3.2 How many litres of water can the tank hold? (1 m³ = 1 000 ℓ) (2)

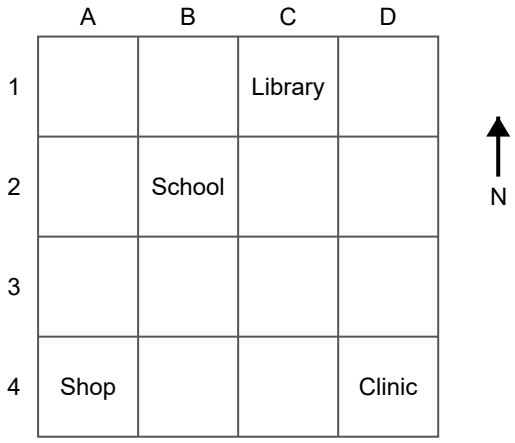
2.4 A bus leaves Polokwane at 06:45. The trip takes 2 hours 50 minutes. At what time does the bus arrive? (2)

2.5 Convert 25 °C to degrees Fahrenheit. Use: °F = (1,8 × °C) + 32 (2)

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QUESTION 3: MAPS AND PLANS

A town map has a scale of 1 : 50 000. The simple grid map below shows four places in the town.



3.1 Explain what the scale 1 : 50 000 means. (2)

3.2 On the map, the school and the clinic are 4,2 cm apart. Calculate the real distance in kilometres. (3)

3.3 A learner walks this distance at 5 km per hour. Calculate how long the walk takes, in minutes. Use: Time = distance ÷ speed (3)

3.4 Give the grid reference of the clinic. (1)

3.5 What is found in block A4? (1)

3.6 In which general direction must the learner walk from the school to reach the clinic? (2)

3.7 A house plan is drawn with a scale of 1 : 100. A bedroom measures 3,5 cm by 4 cm on the plan. Calculate the real length and width of the bedroom in metres, and its real area. (4)

3.8 A learner says: "A map with a scale of 1 : 50 000 shows more detail than a map with a scale of 1 : 250 000." Is the learner correct? Explain. (2)

3.9 Give ONE reason why a map must always have a scale. (2)

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QUESTION 4: DATA HANDLING AND PROBABILITY

The test marks (out of 50) of 11 learners are: 23 35 41 28 35 47 19 35 30 44 27

4.1 Arrange the marks in ascending order. (2)

4.2 Write down the mode. (1)

4.3 Determine the median. (2)

4.4 Calculate the mean (average) mark. (3)

4.5 Calculate the range. (2)

4.6 The pass mark is 50%. How many learners failed, and what percentage of learners passed? (5)

4.7 Which measure, the mean or the median, better describes a typical mark? Give a reason. (2)

4.8 A survey asked 40 learners how they travel to school: Walk 18, Taxi 12, Bus 6, Car 4.

4.8.1 Draw a bar graph to show this information. Give your graph a heading and label the axes. (5)

4.8.2 Calculate the percentage of learners who walk to school. (2)

4.8.3 A learner is chosen at random. What is the probability that the learner travels by taxi? Give your answer as a decimal. (2)

4.9 Explain the difference between discrete and continuous data. Give ONE example of each. (2)

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TOTAL: 100

END OF QUESTION PAPER